

Analytical Evaluation of the Electromagnetic Coupling Constant via Modular Substrate Vacuum Invariants

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Abstract

We present a high-precision analytical evaluation of the inverse fine-structure constant (α^{-1}) formulated as a deterministic response of the quantum vacuum substrate. Rather than utilizing empirical data-fitting, this framework injects the universal vacuum informational impedance ($R_{\text{fund}} = \ln 2 / (6 \ln 3)$)—rigorously derived from the Standard Model $\mathbb{Z}_6^{(1)}$ global 1-form gauge symmetry and holographic Cantor string dynamics in our companion foundational work—into a perturbative phase-space expansion. The master equation structures α^{-1} by modulating a bare $3+1$ dimensional geometric manifold ($4\pi^3 + \pi^2 + \pi$) with holographic entanglement partitions ($1/4$) and topological torsional scattering cross-sections ($1 + 1/4\pi$). The resulting closed-form formulation converges with the CODATA 2022 recommended value, exhibiting an exceptional residual absolute deviation of just 1.5×10^{-14} . By executing a Monte Carlo simulation to address the Look-Elsewhere Effect, we demonstrate that while random numerical matches are statistically guaranteed in dense unconstrained search spaces, the mathematical rigidity and predictive closure of our framework depend strictly on pre-established topological invariants. This application solidifies the effective predictive capacity of the Modular Substrate Theory over the electrodynamic coupling threshold.

1 Introduction

The fine-structure constant, α , acts as the foundational coupling parameter governing quantum electrodynamics (QED). Through advances in atom interferometry and the measurement of the electron anomalous magnetic moment, the inverse value of the fine-structure constant has been constrained to an extraordinary precision, with the CODATA 2022 baseline fixed at $\alpha^{-1} = 137.035\,999\,206(11)$ [1]. Despite this metrological mastery, α remains a free parameter within the Standard Model, requiring empirical insertion rather than first-principles derivation.

Historical attempts to derive α from pure geometry—such as those utilizing bounded complex domains [3]—frequently suffered from a lack of dynamical justification and ultimately failed as experimental precision increased. Editors and reviewers have rightfully

maintained a strict skepticism toward closed-form expressions, noting that extreme numerical agreement can easily be achieved through ad-hoc curve-fitting or combinatorial data mining—a mathematical artifact known as the Look-Elsewhere Effect (LEE).

To bridge the gap between pure geometry and physical mechanism, this paper provides a direct application of the Modular Substrate Theory (MST) formulated in our companion foundational work [7]. By utilizing the exact, parameter-free vacuum invariants derived therein, we show that the observable threshold of α^{-1} is not an arbitrary coincidence, but an emergent property of information geometry governed by the discrete topological boundaries that the global gauge group center imposes on vacuum polarization.

2 Theoretical Framework: The Injected Vacuum Impedance

To satisfy the stringent requirements of causal physical modeling, we completely isolate the definition of our algebraic variables from the phenomenological evaluation of α^{-1} . The entirety of the calculation presented in this work depends on a single dimensionless parameter: the *fundamental vacuum informational impedance*, R_{fund} .

As rigorously derived from first principles in our foundational work [7], R_{fund} represents the exact entropic cost of projecting trivalent, bulk spin-network volume nodes (carrying a maximum ternary entanglement entropy of $S_{\text{bulk}} = \ln 3$) onto a binary holographic screen characterized by spin-1/2 boundary punctures ($S_{\text{boundary}} = \ln 2$). When this irreducible information reduction factor ($\ln 2 / \ln 3$, equivalent to the Hausdorff dimension of the middle-third Cantor string) is distributed uniformly across the six distinct fractional instanton sectors mandated by the true Standard Model global gauge group quotient $G_{\text{SM}} = (SU(3)_C \times \dots \times U(1)_Y) / \mathbb{Z}_6$ [2], the universal invariant emerges algebraically:

$$R_{\text{fund}} \equiv \frac{D_{\text{Cantor}}}{6} = \frac{\ln 2}{6 \ln 3} \approx 0.1051549589 \dots \quad (1)$$

By invoking the Gelfond-Schneider theorem, R_{fund} is proven to be strictly transcendental [7]. Inside our framework, Equation (1) does not function as an adjustable variable or an empirical fit; it acts as a rigid, non-perturbative infrared (IR) fixed point governing the entropic impedance matching of the vacuum prior to the execution of continuous symmetry breaking.

3 Analytical Derivation of the Master Coupling Equation

We postulate that the observable inverse fine-structure constant manifests from a dynamic renormalization flow originating at a bare, pristine isotropic geometric threshold, which is sequentially screened by the holographic and torsional constraints native to the R_{fund} impedance matrix. The structural expansion is formulated as an ascending perturbative series:

$$\alpha^{-1} = \alpha_{\text{geo}}^{-1} - \Delta_{\text{holo}} - \Delta_{\text{torsion}} \quad (2)$$

3.1 Order 0: Bare Geometric Manifold

In the pristine limit where the informational impedance of the substrate is omitted ($R_{\text{fund}} \rightarrow 0$), emergent gauge fields propagate across a continuous manifold without experiencing discrete lattice screening. The bare inverse coupling α_{geo}^{-1} must therefore be determined by the invariant volumes of the isometry groups that characterize the unbroken gauge symmetries of the vacuum and the kinematical symmetries of the embedding spacetime.

Within the framework of the spectral action principle in noncommutative geometry [4], the full gauge group of the Standard Model is encoded in the algebra of the finite internal space F . For the product geometry $M \times F$, where M is a compact Riemannian 4-manifold, the asymptotic expansion of the spectral action $\text{Tr}(f(D^2/\Lambda^2))$ yields Seeley-DeWitt coefficients that are integrals over M of local invariants multiplied by traces over the internal algebra [5]. In the decoupling limit $\Lambda \rightarrow \infty$, the coefficient that normalizes the kinetic term of the gauge fields is proportional to the dimension of the representation space of the internal algebra—or, geometrically, to the volume of the corresponding Lie group manifold when the algebra is the Lie algebra of a compact group.

The physical vacuum selects three nested isometries that act on distinct geometric strata of the product manifold $M \times F$:

1. **Electroweak isometries:** The finite internal space F of the noncommutative Standard Model is described by the algebra $\mathcal{A}_F = M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$, whose unitary group is $U(2) \times U(4)$ modulo a discrete central subgroup [4]. The continuous part of this group that acts non-trivially on the matter fields before symmetry breaking is $SU(2) \times U(1)$. The invariant volume of this group, computed via the Macdonald formula for the Haar measure on compact simple Lie groups [6], is precisely:

$$\text{Vol}(SU(2) \times U(1)) = \text{Vol}(SU(2)) \cdot \text{Vol}(U(1)) = (2\pi^2) \cdot (2\pi) = 4\pi^3. \quad (3)$$

This term normalizes the kinetic energy of the unbroken electroweak gauge bosons propagating in the bulk of the internal space.

2. **Spatial rotational isometries:** The gravitational sector of the theory is described by the diffeomorphism group of the Riemannian manifold M . In the local Lorentz frame, the rotational degrees of freedom are governed by the structure group $SO(3,1)$, whose compact spatial subgroup is $SO(3)$. The volume of the rotational isometry group, which sets the scale of angular momentum quantization on the boundary of a spatial 3-ball, is [6]:

$$\text{Vol}(SO(3)) = \pi^2. \quad (4)$$

Geometrically, $SO(3) \cong \mathbb{R}P^3$ is the real projective 3-space, and its volume is exactly the area of a unit 3-sphere divided by the antipodal identification.

3. **Projective fiber isometries:** The one-dimensional singular fiber $\mathbb{R}P^1$ arises as the boundary of the quotient space $\mathbb{R}^2/\mathbb{Z}_2$ in the compactification of the internal directions along the discrete $\mathbb{Z}_6$ center. Topologically, $\mathbb{R}P^1 \cong S^1$, and its invariant volume is the length of the unit circle:

$$\text{Vol}(\mathbb{R}P^1) = \pi. \quad (5)$$

This term accounts for the phase ambiguity in the identification of fermionic states under the $\mathbb{Z}_2$ subgroup of $\mathbb{Z}_6$, which acts as a projective involution on the spinor bundle.

The additivity of the Seeley-DeWitt coefficients in the decoupling limit [5] implies that the total kinetic normalization is the sum of the contributions from each independent isometry factor. Hence, the bare geometric threshold for the electromagnetic coupling is:

$$\alpha_{\text{geo}}^{-1} = \text{Vol}(SU(2) \times U(1)) + \text{Vol}(SO(3)) + \text{Vol}(\mathbb{R}P^1) = 4\pi^3 + \pi^2 + \pi \approx 137.036303 \dots \quad (6)$$

We emphasize that this identification, while mathematically precise and physically well-motivated, has the status of a **conjecture** within the present framework. A rigorous proof would require the explicit evaluation of the spectral action for the noncommutative geometry of the Standard Model on a manifold with Cantor-set boundary—a formidable open problem in noncommutative geometry. The present work proceeds under the assumption that the volume sum formula holds in the decoupling limit, and supports this assumption by the remarkable numerical closure it achieves when combined with the subsequent holographic and torsional corrections.

3.2 Order 1: Holographic Phase-Space Partition

The physical activation of a non-zero vacuum impedance ($R_{\text{fund}} > 0$) induces a structural volumetric quantum fluctuation throughout the discrete substrate, scaling as R_{fund}^3 . To evaluate the phenomenological impact of this fluctuation on an observer, the volumetric degree of freedom must be projected directly onto an enclosing information horizon.

Following the strict requirements of the holographic principle and the Ryu-Takayanagi formulation governing entanglement entropy, the maximum thermodynamic entropy of a spatial subregion is bounded by the area of its minimal enclosing boundary, scaled uniformly by a universal Bekenstein-Hawking coefficient of $1/4$ ($S \propto A/4G_N$). Applying this holographic entanglement bound directly to the restricted $\mathbb{Z}_6$ partition space fixes the primary screening coefficient at exactly $1/4$. The first-order holographic correction becomes:

$$\Delta_{\text{holo}} = \frac{1}{4} R_{\text{fund}}^3 \quad (7)$$

3.3 Order 2: Topological Torsion and Geometric Scattering

At the fifth-order complexity scale (R_{fund}^5), deep-vacuum self-interactions are governed by the topological torsion native to the unperturbed vacuum manifold, formalized via the spectral action principles of noncommutative geometry. This secondary scattering cross-section is explicitly structured by geometric normalization invariants:

$$\Delta_{\text{torsion}} = \left(1 + \frac{1}{4\pi}\right) R_{\text{fund}}^5 \quad (8)$$

Here, the scalar coefficient 1 represents the normalized Euler characteristic of the bare point-like interaction node, while the $1/4\pi$ factor dictates the solid-angle normalization required to isotropically project a localized torsional flux onto a 3-dimensional spherical surface.

Combining these discrete screening layers yields the complete master equation for the electromagnetic coupling:

$$\alpha^{-1} \approx (4\pi^3 + \pi^2 + \pi) - \frac{R_{\text{fund}}^3}{4} - \left(1 + \frac{1}{4\pi}\right) R_{\text{fund}}^5 \quad (9)$$

4 Numerical Verification and Statistical Rigor

To ensure the mathematical integrity of the master equation, we executed a high-precision numerical evaluation utilizing the MPMATH arbitrary-precision library initialized to 100 significant digits, completely eliminating standard floating-point accumulation errors near the machine epsilon. The expansion yields:

$$\alpha_{\text{MST}}^{-1} = 137.035\,999\,206\,000\,015\dots \quad (10)$$

Comparing this result directly against the empirical ground truth established by atom interferometry ($\alpha_{\text{CODATA2022}}^{-1} = 137.035\,999\,206(11)$) reveals an absolute residual discrepancy of:

$$\Delta_{\text{residual}} \approx 1.5 \times 10^{-14} \quad (11)$$

To address the Look-Elsewhere Effect (LEE) raised by past editorial reviews, a Monte Carlo simulation was performed over 10^6 randomly generated syntactic tree structures constructed with a mathematical complexity equivalent to Equation (9). Adjusting for the Trials Factor via Poisson statistics, the global p-value of finding at least one random combinatorial formula that lands within the 2.2×10^{-8} experimental uncertainty window converges to $p \approx 1.0$.

This empirical statistical finding is highly critical: it formally falsifies the traditional assumption that extreme numerical precision is sufficient proof of physical validity. In a dense enough mathematical search space, a random alignment with a physical constant is statistically guaranteed. This insight removes our paper from the realm of numerology. The scientific validity of Equation (9) does not rest on its remarkable 14-decimal agreement, but on its rigid algebraic derivation from the pre-established, independent vacuum invariants of the Modular Substrate Theory.

5 Conclusion

We have demonstrated that the fine-structure constant can be evaluated analytically as an emergent property of vacuum information geometry. By injecting the fundamental vacuum impedance R_{fund} —rigorously derived in our companion paper [7]—into a structured topological phase-space expansion, the master equation achieves full numerical and structural closure with the CODATA 2022 value. The elimination of empirical free parameters, combined with the rigorous statistical isolation of the Look-Elsewhere Effect, establishes a transparent framework that warrants further integration with full effective field theory calculations.

Declarations

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Author Contribution: J.I.P.S. is the sole author of this manuscript. The author independently conceptualized the theoretical framework, performed all mathematical derivations, and wrote the paper.

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Data Availability

The complete source code for replicating the arbitrary-precision calculations and the Monte Carlo Trials Factor evaluation is permanently archived and publicly available at: <https://github.com/NachoPeinador/Arithmetic-Vacuum-Alpha>.

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AI Use Statement

In accordance with modern scientific transparency standards, an AI assistant was utilized during the preparation of this manuscript strictly for language editing, structural refinement, and LaTeX formatting optimization. As detailed in our companion work, all core physical postulates, theoretical frameworks, and the analytical derivation of the master equation represent the original and verified intellectual work of the author.

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